

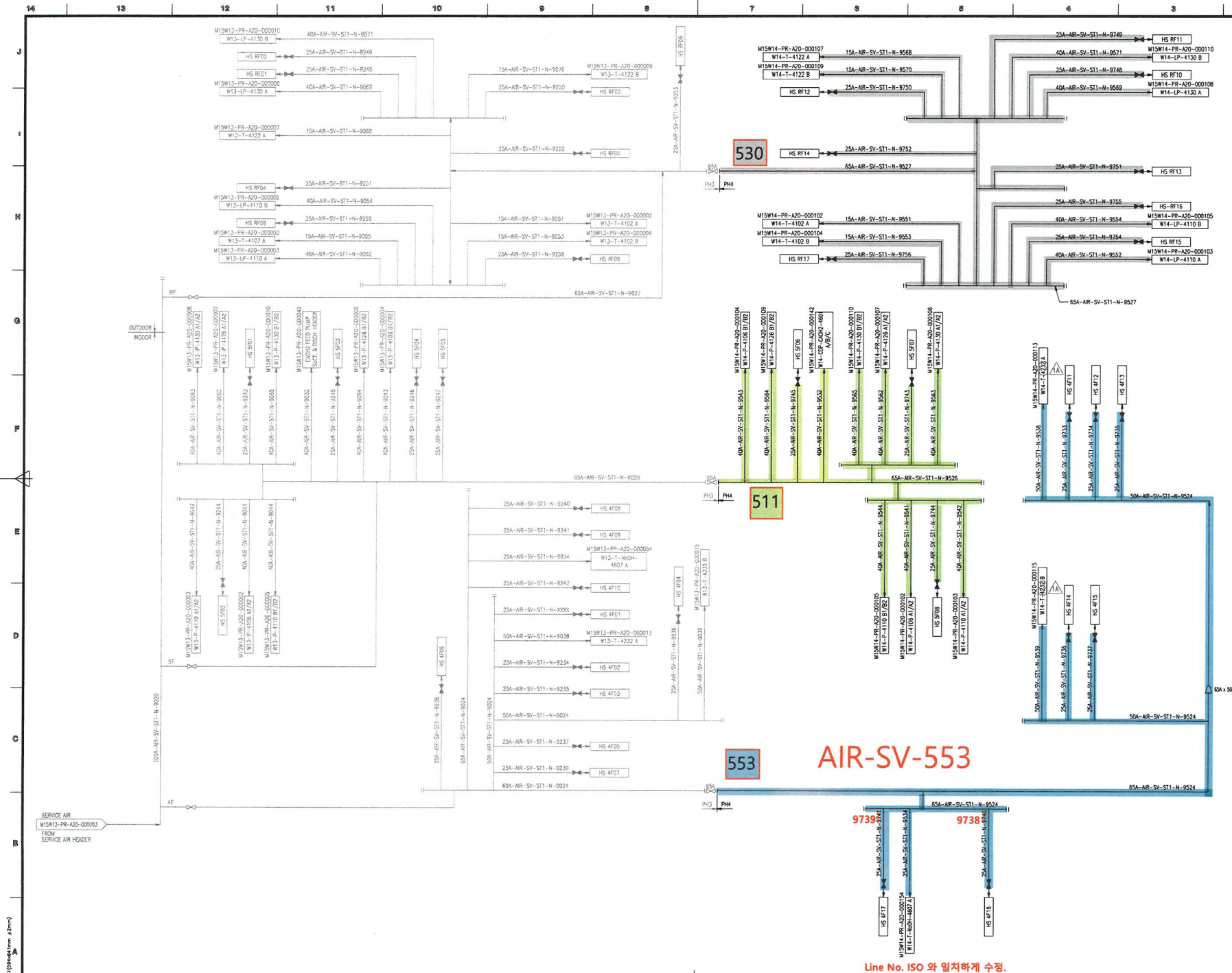
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

AIR-SV-553

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	553	4FA	25A-AIR-SV-ST1-N-9534-01	1 OF 1	1	CJ-M15W14-PR-A20-000154	2A	
2	553	4FA	25A-AIR-SV-ST1-N-9598-01	1 OF 1	1	CJ-M15W14-PR-A20-000154	2A	
3	553	4FA	25A-AIR-SV-ST1-N-9733-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
4	553	4FA	25A-AIR-SV-ST1-N-9734-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
5	553	4FA	25A-AIR-SV-ST1-N-9735-01	1 OF 1	1	CJ-M15W13-PR-A20-000083-A	2	
6	553	4FA	25A-AIR-SV-ST1-N-9736-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
7	553	4FA	25A-AIR-SV-ST1-N-9737-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
8	553	4FA	25A-AIR-SV-ST1-N-9738-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
9	553	4FA	25A-AIR-SV-ST1-N-9739-01	1 OF 1	0	CJ-M15W13-PR-A20-000083-A	2	
10	553	4FA	50A-AIR-SV-ST1-N-9538-01	1 OF 1	1	CJ-M15W14-PR-A20-000113	2A	
11	553	4FA	✓ 50A-AIR-SV-ST1-N-9539-01	1 OF 1	1	CJ-M15W14-PR-A20-000115	2A	
12	553	4FA	65A-AIR-SV-ST1-N-9524-01	1 OF 4	0	CJ-M15W13-PR-A20-000083-A	2	
13	553	4FA	65A-AIR-SV-ST1-N-9524-02	2 OF 4	1	CJ-M15W13-PR-A20-000083-A	2	
14	553	4FA	65A-AIR-SV-ST1-N-9524-03	3 OF 4	0	CJ-M15W13-PR-A20-000083-A	2	
15	553	4FA	65A-AIR-SV-ST1-N-9524-04	4 OF 4	0	CJ-M15W13-PR-A20-000083-A	2	



대외비

KEY PLAN

NOTE

1A	REVISED A/C	25.08.14
1	REVISED A/C	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
주요 REV.	설계 DESCRIPTION	날짜 DATE
제 DRAWN BY	이인학	
검 CHECKED BY	한정학	
승 APPROVED BY	박동명	
SK 에코플랜트		
공시명 PROJECT TITLE		
M15X Ph4 WWT Project		
PROJECT NO. 223300		
2	REVISED A/C	24.06.21
1	REVISED A/C	23.02.28
0	APPROVAL FOR CONSTRUCTION	23.09.23
주요 REV.	설계 DESCRIPTION	날짜 DATE
제 DRAWN BY	한정학	
검 CHECKED BY	심명용	
승 APPROVED BY	박동명	
SK 에코플랜트		
공시명 PROJECT TITLE		
M15X Ph3 WWT Project		
도면명 DWG. TITLE		
UTILITY DISTRIBUTION DIAGRAM SERVICE AIR DISTRIBUTION SYSTEM (2/2)		
주 SCALE		
NONE		
날 DATE		
2024. 06. 21		
PROJECT NO. 223300		REV.
도면번호 SHEET NO. CJ-M15W13-PR-A20-000083-A		2

ITEM NO.	W14-T-NaOH-4607 A/B
SERVICE	NaOH STORAGE TANK
QTY	2
TYPE	VERTICAL CYLINDRICAL
SIZE	Ø x H m 3.6 x 4.5 (EH4.2)
CAPA.	m3 46 (EH4.3) 1A
MAT'L	FRP

CONTROL
W15W14-PR-A20-000155
TO NaOH FEED PUMP

SIGNAL
W15W14-PR-A20-000318
TO REUSE NaOH FEED PUMP
W14-CDP-NaOH-4707A/B,
4759A/B, 4759B/B

DOOR
W15W14-PR-A20-000054
TO VENT HEADER

CONTROL
W15W14-PR-A20-000060
TO CHEMICAL UNLOADING STATION
W13-CTP-NaOH-4673

NaOH
W15W14-PR-A20-000060
FROM NaOH UNLOADING PUMP
W13-CTP-NaOH-4673

SERVICE AIR
W15W14-PR-A20-000063
FROM LUD FOR SERVICE AIR

AIR-SV-553

25A-AR-SV-ST1-N-9534

00110-NaOH-HP2-N-6503

00110-NaOH-HP2-N-6504

00110-NaOH-HP2-N-6505

00110-NaOH-HP2-N-6506

00110-NaOH-HP2-N-6507

00110-NaOH-HP2-N-6508

00110-NaOH-HP2-N-6509

00110-NaOH-HP2-N-6510

00110-NaOH-HP2-N-6511

00110-NaOH-HP2-N-6512

00110-NaOH-HP2-N-6513

00110-NaOH-HP2-N-6514

00110-NaOH-HP2-N-6515

00110-NaOH-HP2-N-6516

00110-NaOH-HP2-N-6517

00110-NaOH-HP2-N-6518

00110-NaOH-HP2-N-6519

00110-NaOH-HP2-N-6520

00110-NaOH-HP2-N-6521

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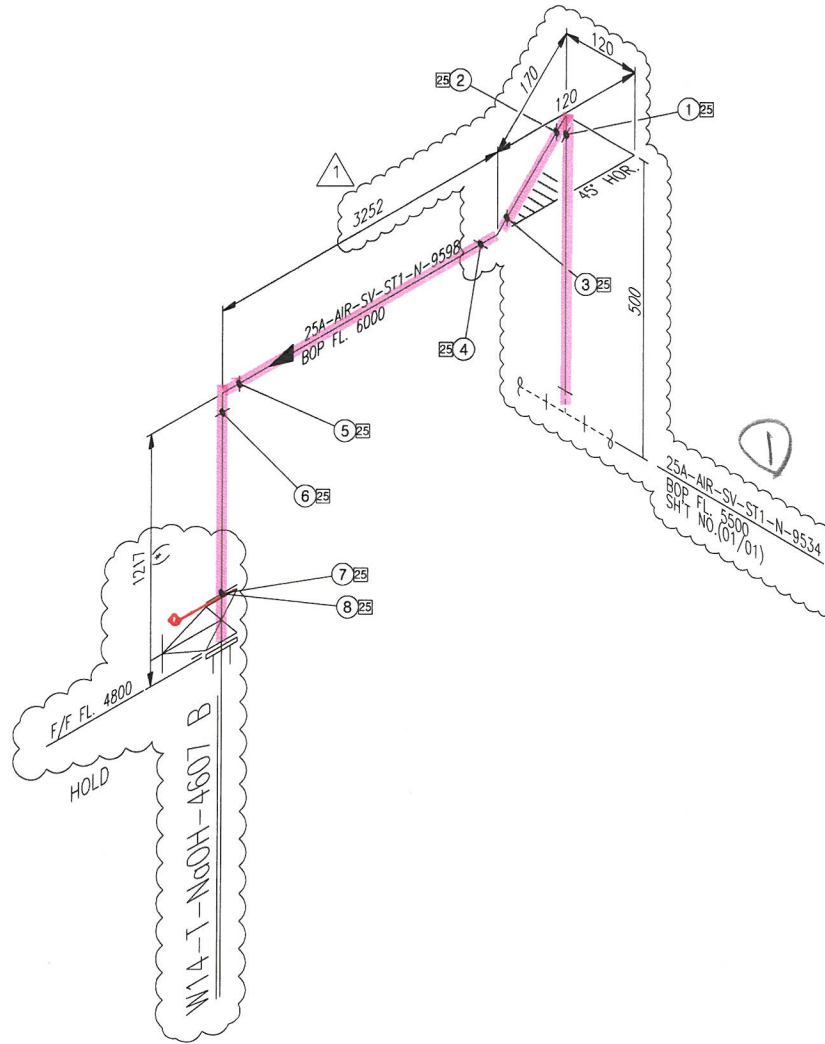
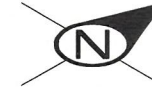
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00110-NaOH-HP2-N-6707

00110-NaOH-HP2-N-6708

00110-NaOH-HP2-N-6709



<NOTE>
1. (*) FIELD ADJUST

SK ecoplant

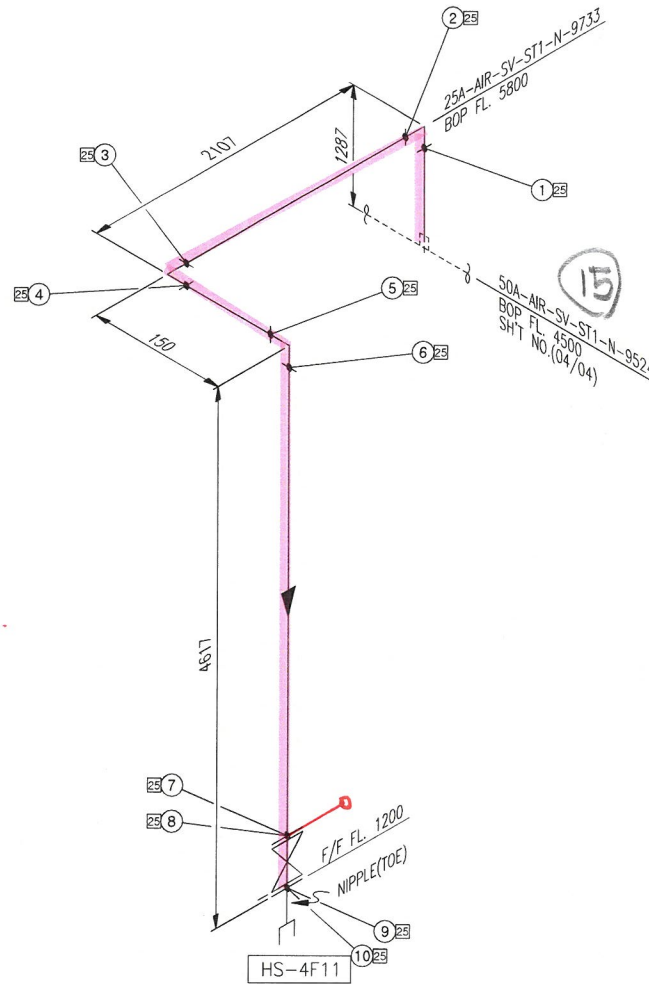
ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	25A	5.1M	PIPE, STS 304TP, ERW, PE - SCH 10S
E90	25A	2EA	ELBOW 90 DEG, STSF304, SW - SCH 10S
E45	25A	1EA	ELBOW 45 DEG, STSF304, SW - SCH 10S
VG	25A	1EA	GATE VALVE, KS 10K, RF FLC'D, BODY-STSF304 / TRIM-STIS304, BB / OS&Y, HANDWHEEL
F	25A	1EA	FLG SO, KS 10K, RF, STIS304
G	25A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	25A	2SET	MACHINE BOLT & NUTS : STS 304 / W WASHERS

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CONFLD : RC * ECC SWAGE : RE * CON SWAGE : RCS/RCS1 * ECC SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PPVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB10/VB2/VB210 * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD10 * DAMPER : VP * QUICK COUPLING : SOC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14
	THICK.	-	DESIGN	10	60	UNIT	4FA
			OPERATING	7.5	AMB	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-040009-D	
						REF. P & ID	
						CJ-M15W14-PR-A20-000154	

CLIENT	SK hynix	PROJECT NAME	M15X Ph4 Project
CONTRACTOR	SK ecoplant	JOB NO.	22330D
COMMODITY	AIR-SV	SH'T NO.	01 / 01
LINE NO.	25A-AIR-SV-ST1-N-9598	REV.	1

경고 : 본 도면은 SK HYNIX의 기밀파일 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



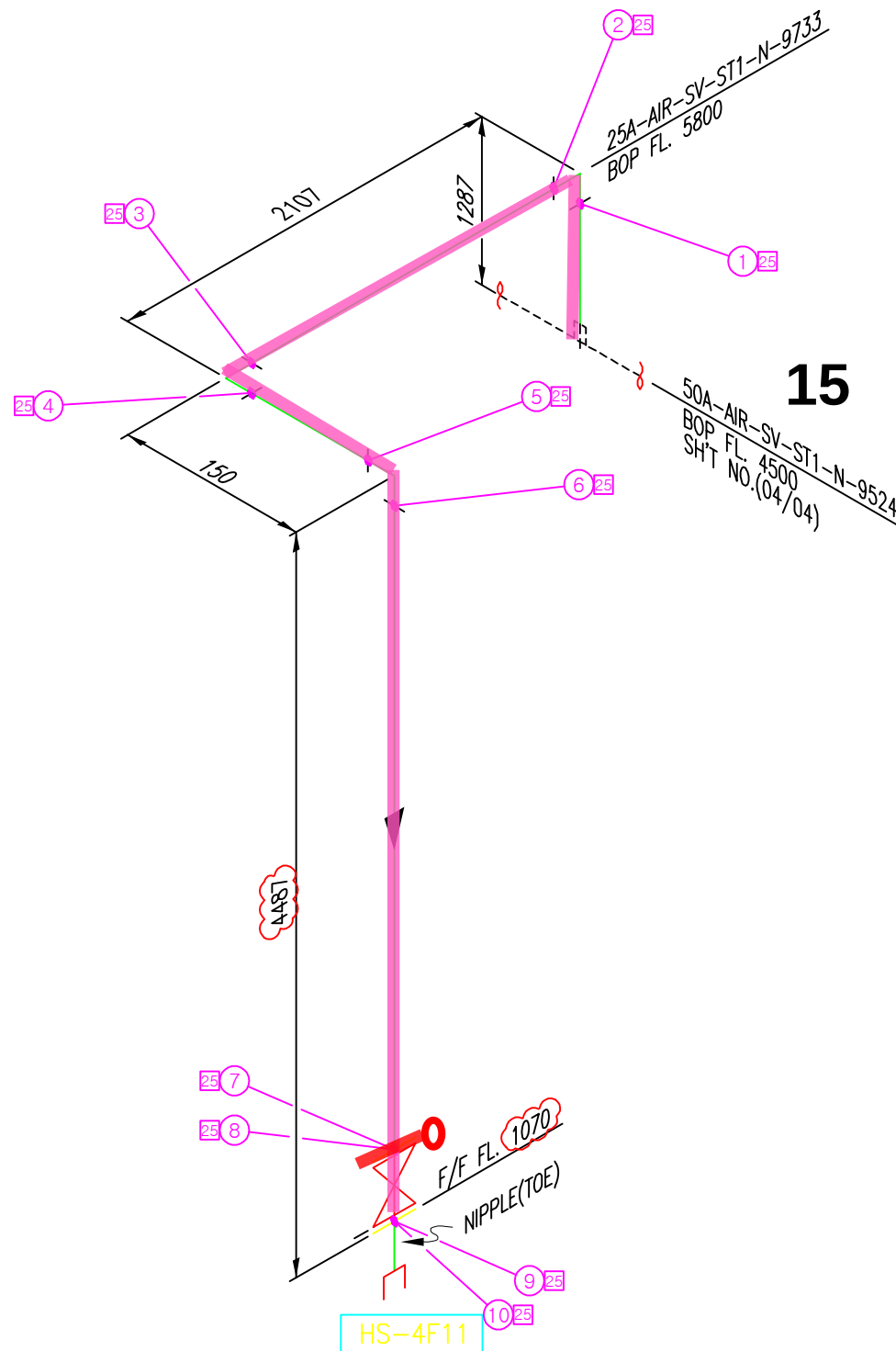
1. (*) FIELD ADJUST

[illegible]

③

[illegible]

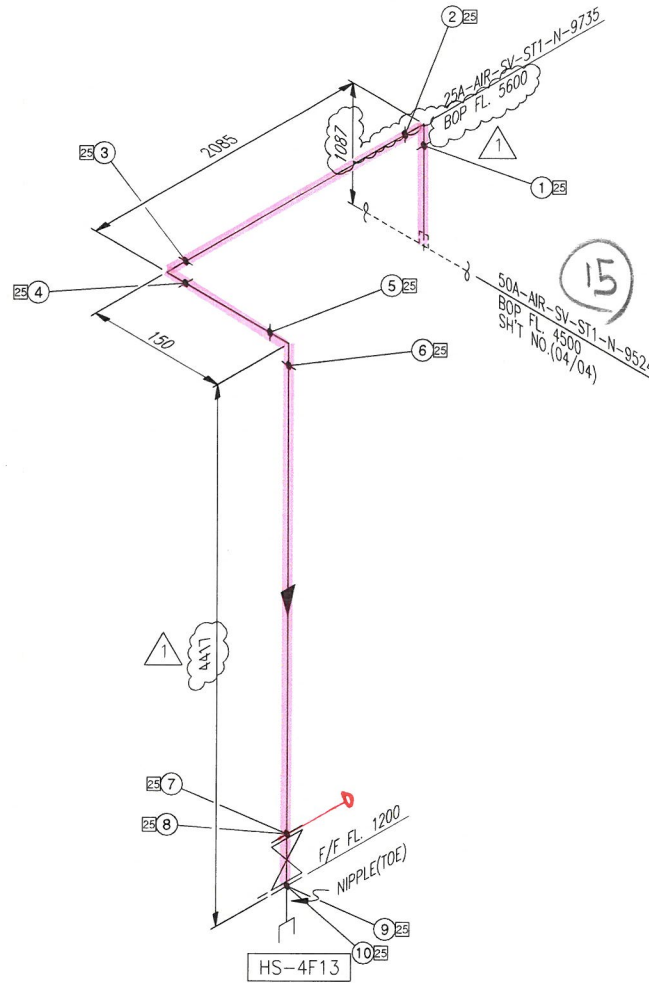
경고 : 해당 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 복제 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생한 경우 유표자에게 민형사상 책임을 요구할 수 있음.



1. (*) FIELD ADJUST

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	-	DESIGN	10	60	UNIT	4FA						
			OPERATING	7.5	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-040001-C			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
						REF. P. & ID							
						CJ-M15W13-PR-A20-000083-A		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

CLIENT		3	
			
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
AIR-SV		22330D	
LINE NO.	25A-AIR-SV-ST1-N-9733	SH'T NO.	REV.
		01 / 01	0

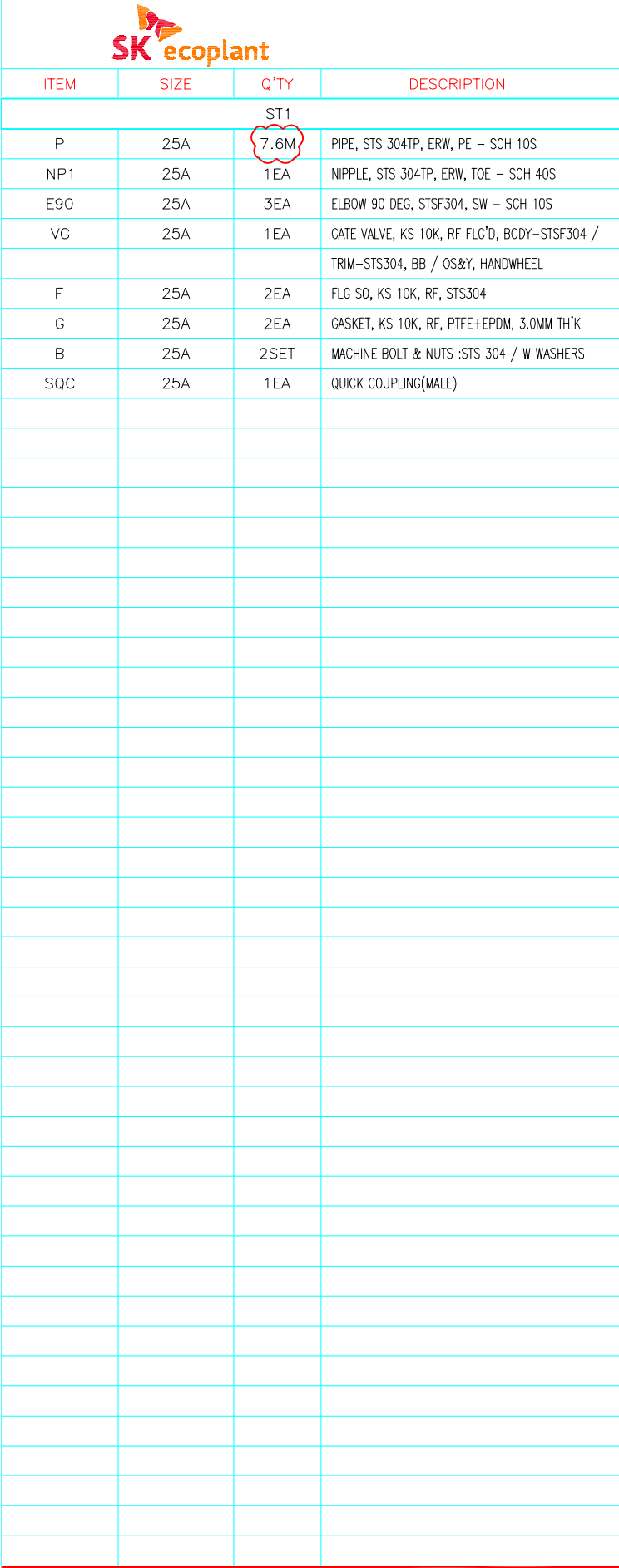


<NOTE>
1. (*) FIELD ADJUST

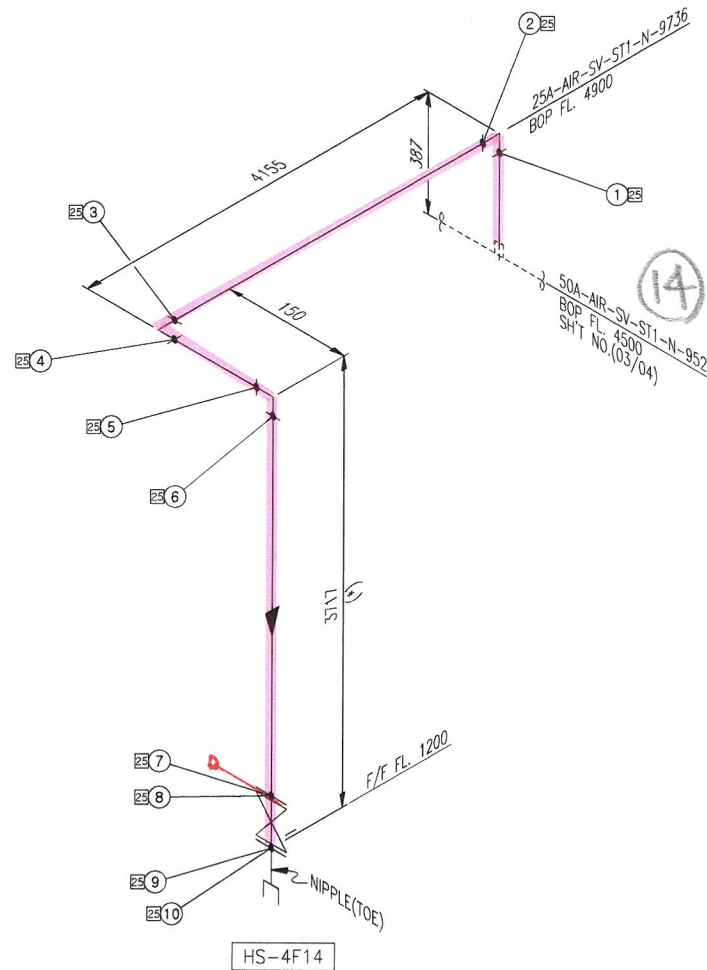
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HOPE/PMC) : SC * SADDLE(HOPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VCS/VCS1/VCS2
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SOC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14					CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	10	60	UNIT	4FA					SK ecoplant	M15X Ph4 Project
			OPERATING	7.5	AMB	REF. PLAN DWG.						COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-040002-C						AIR-SV	22330D
						REF. P & ID						LINE NO.	SH'T NO. REV.
						CJ-M15W13-PR-A20-000083-A						25A-AIR-SV-ST1-N-9735	01 / 01 1
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



CLIENT	 	5
CONTRACTOR		PROJECT NAME M15X Ph4 Project
COMMODITY	AIR-SV	JOB NO. 22330D
LINE NO.	25A-AIR-SV-ST1-N-9735	SH'T NO. 01 / 01 REV. 1



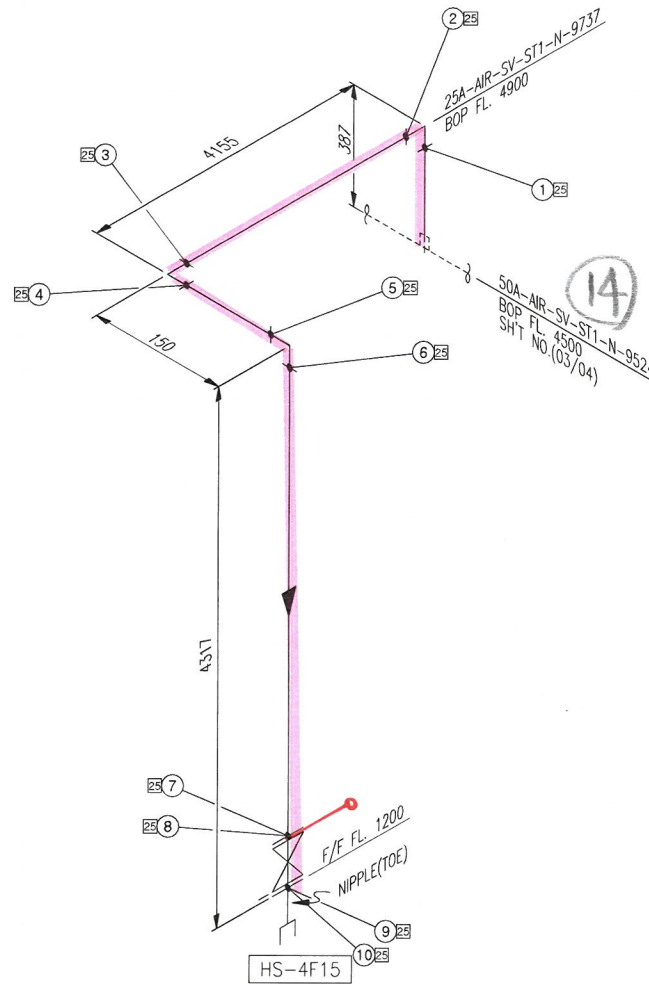
* PIPE: P/P1 * 90° ELBOW: E90 * 45° ELBOW: E45 * TEE: T * RED.TEE: TR * STUB-IN: STU * REINFORCING PAD: RP * CON.RED: RC * COC.RED: RE * CON.SWAGE: RCS/RCS1 * ECC.SWAGE: RES/RES1 * NIPPLE: NP/NP1/NP2 * CAP: CAP/CAP1 * PLUG: PLG * FULL COUPLING: FC * HALF COUPLING: HF/HF1 * STUB-IN: STU * SOCKET(HOPE/PC): SC * SADDLE(HOPE): SD * FLANGE: F/F1 * GASKET: G/G1/G2 * BOLT & NUT: B/B1 * BLIND FLANGE: FB/FB1 * GATE VA.: VG/VG1/VG2 * BALL VA.: VB/VB0/VB1/VB10/VB2/VB20 * BUTTERFLY VA.: VF/VF4 * CHECK VA.: VC/VC1/VC2/VC3/VC4/VCS/VCS1/VCS2/VCS3/VCS4/VCS5/VCS6/VCS7 * GLOBE VA.: VL/VL1/VL2 * DIAPHRAGM VA.: VD/VD0 * DAMPER: VP * QUICK COUPLING: SQC * FLEXIBLE JOINT(STEEL): FLXB * FLEXIBLE JOINT(RUBBER): FLXR * FLEXIBLE JOINT(STEEL PTFE LINED): FLXL * FLEXIBLE HOSE: FLXH * STEAM TRAP: ST * SPRAY NOZZLE: SPN * EMERGENCY SHOWER: SSH * STRAINER(Y-TYPE): YSTR * STRAINER(T-TYPE): TSTR * STRAINER(BUCKET-TYPE): BSTR * ANCHOR PLATE: STOP * BIRD SCREEN: BS * ISOLATION KIT: IK * UNLOADING BOX: UB * AIR TRAP: AT * LOTO: LOTO * SIGHT GLASS: SG * PRESSURE GAUGE: PG * SHIELD: SH * SPECIAL BOLT & NUT: BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (℃)	AREA W14											CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	10	60	UNIT 4FA												M15X Ph4 Project
			OPERATING	7.5	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040004-C											COMMODITY	JOB NO.
						REF. P & ID CJ-M15W13-PR-A20-000083-A		25.01.24	APPROVED FOR CONSTRUCTION	황재홍	안남후	박동영						22330D
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D						LINE NO.
																	25A-AIR-SV-ST1-N-9736	.01 / 01 0

경고 : 해당 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민·형사상 책임을 요구할 수 있음.

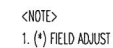
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CLIENT	 	
CONTRACTOR		PROJECT NAME M15X Ph4 Project
COMMODITY	AIR-SV	JOB NO. 22330D
LINE NO.	25A-AIR-SV-ST1-N-9736	SH'T NO. 01 / 01 REV. 0



* PIPE: P/P1	* 90° ELBOW: E90	* 45° ELBOW: E45	* TEE: T	* RED TEE: TR	* STUB-IN: STU	* REINFORCING PAD: RP	* CON. RED: RC	* ECC. RED: RE	* CON. SWAGE: RCS/RCS1	* ECC. SWAGE: RES/RES1	* NIPPLE: NP/NP1/NP2	* CAP: CAP/CAP1	* PLUS: PLG	* FULL COUPLING: FC	* HALF COUPLING: HF/HF1
* STUB-IN: STU	* SOCKET(HDPE/PC): SC	* SADDLE(HDPE): SD	* FLANGE: F/F1	* GASKET: G/G1/G2	* BOLT & NUT: B/B1	* BLIND FLANGE: FB/FB1	* GATE VA.: VG/VG1/VG2	* BALL VA.: VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA.: VF/VF4	* CHECK VA.: VC/VC1/VC2/VC3/VC4/VCS/VCS1/VCS2/VCS3/VCS4/VCS5/VCS6/VCS7					
* GLOBE VA.: VL/VL1/VL2	* DIAPHRAGM VA.: VD/VD0	* DAMPER: VP	* QUICK COUPLING: SQC	* FLEXIBLE JOINT(STEEL): FLXB	* FLEXIBLE JOINT(RUBBER): FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED): FLXL	* FLEXIBLE HOSE: FLXH	* STEAM TRAP: ST	* SPRAY NOZZLE: SPN	* EMERGENCY SHOWER: SSH					
* STRAINER(TYPE): YSTR	* STRAINER(T-TYPE): TSTR	* STRAINER(BUCKET-TYPE): BSTR	* ANCHOR PLATE: STOP	* BIRD SCREEN: BS	* ISOLATION KIT: IK	* UNLOADING BOX: UB	* AIR TRAP: AT	* LOTO: LOTO	* SIGHT GLASS: SG	* PRESSURE GAUGE: PG	* SHIELD: SH	* SPECIAL BOLT & NUT: BL/BL1			

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 복제와 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사나 유보자의 불법 사용으로 피해가 발생할 경우 유보자에게 민/형사상 책임을 요구할 수 있음.



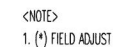
CLIENT



(8)

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
AIR-SV		22330D	
LINE NO.	25A-AIR-SV-ST1-N-9738	SH'T NO.	REV.
		01 / 01	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음.



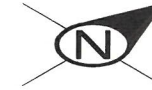
CLIENT



9

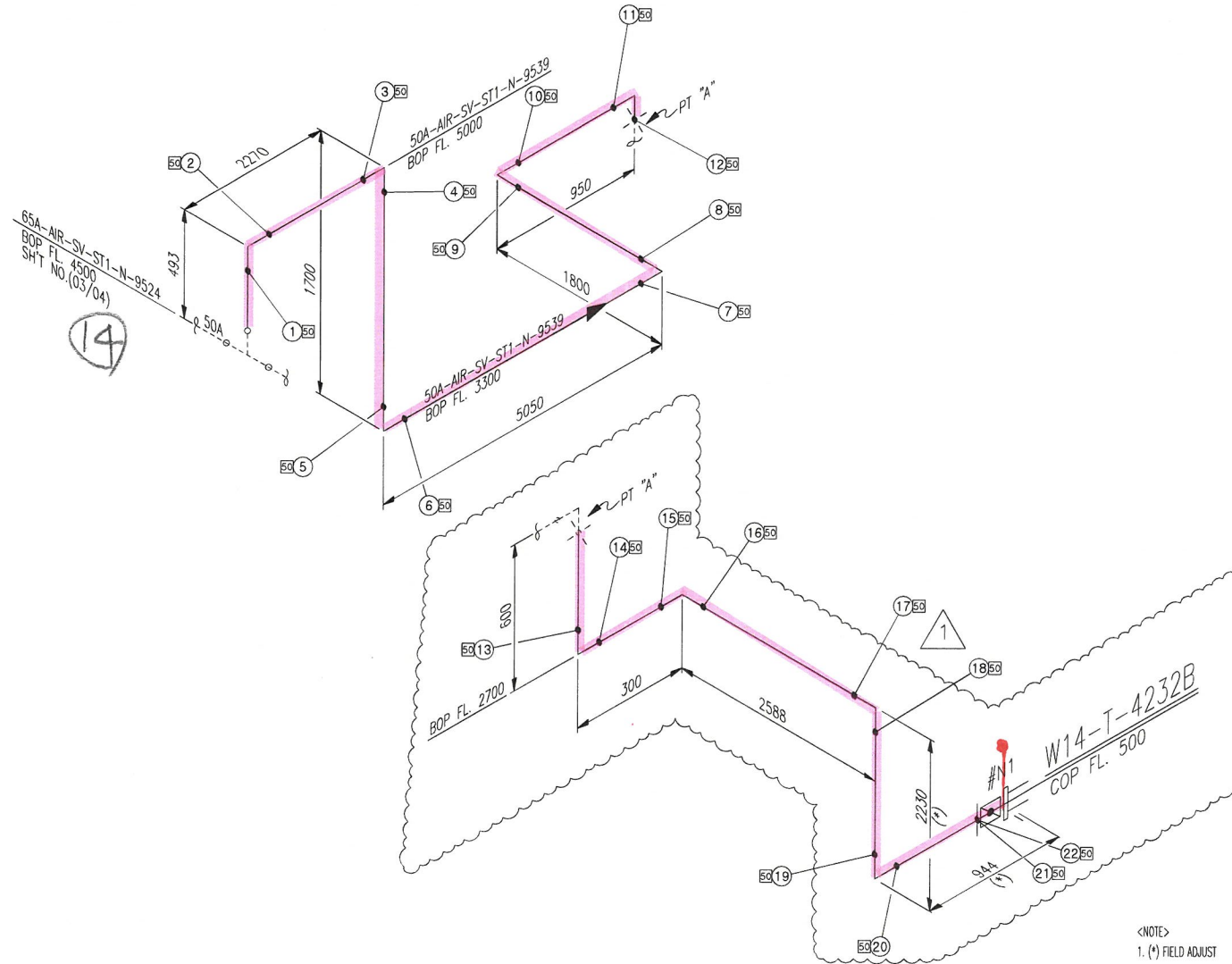
[illegible]

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 복제 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		ST1	
P	50A	18.0M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	10EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
VF	50A	1EA	BUTTERFLY VALVE, KS 10K, WAFER RF, BODY-OC450 / TRIM-STIS304 / EPDM-LINED, LEVER
F	50A	1EA	FLG SO, KS 10K, RF, STS304
G	50A	2EA	GASKET, KS 10K, RF, PIPE+EPDM, 3.0MM THK
BL	50A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)



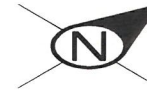
<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BUND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB21 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1 * DAMPER : VP * QUICK COUPLING : SQG * FLEXIBLE JOINT(STEEL) : FLXG * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

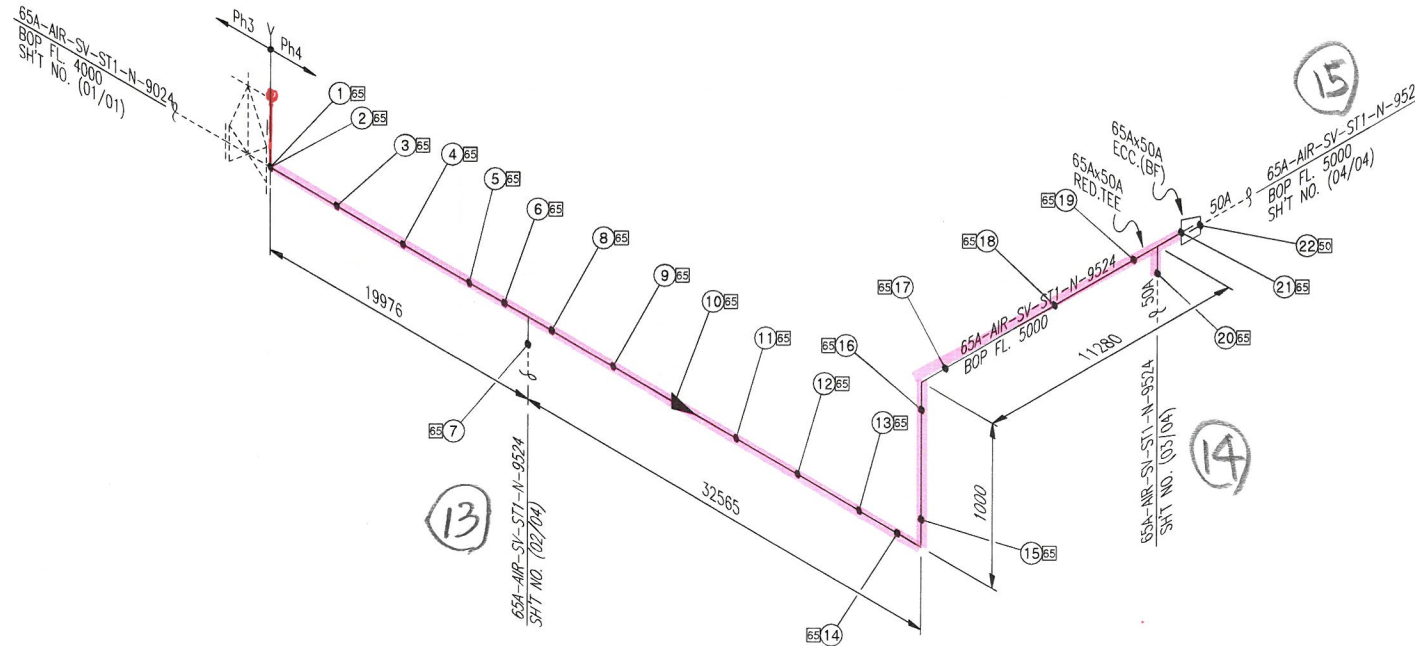
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14
	THICK.	-	DESIGN	10	60	UNIT	4FA
			OPERATING	7.5	AMB	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-040004-C	
						REF. P & ID	
						CJ-M15W13-PR-A20-000083-A	

CLIENT	SK hynix	PROJECT NAME	M15X Ph4 Project
CONTRACTOR	SK ecoplant	JOB NO.	22330D
COMMODITY	AIR-SV	SH'T NO.	01 / 01
LINE NO.	50A-AIR-SV-ST1-N-9539	REV.	1

경고: 본 도면은 SK HYNIX의 기밀파일 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	65A	64.8M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	65A	2EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	65A	1EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
TR	65Ax50A	1EA	TEE RED, STS304, ERW, BE - SCH 10S
RE	65Ax50A	1EA	REDUCER ECC, STS304, ERW, BE - SCH 10S
F	65A	1EA	FLG SO, KS 10K, RF, STS304
G	65A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	65A	1EA	MACHINE BOLT & NUTS :STS 304 / W WASHERS



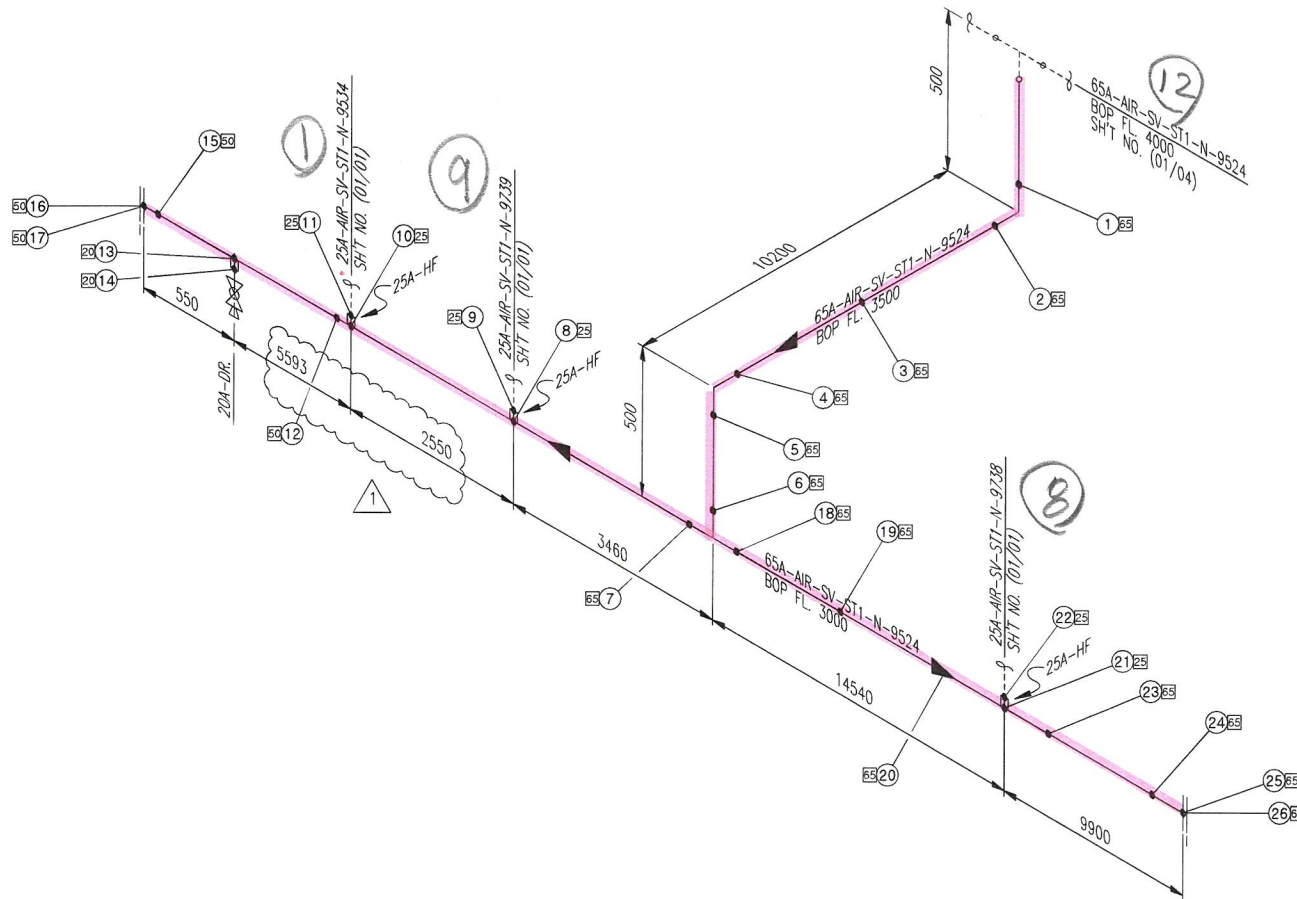
<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FR/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB1.0/VB2/VB2.0 * BUTTERFLY VA : VF/VF1 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD1.0 * DAMPER : VP * QUICK COUPLING : SOC * FLEXIBLE JOINT(STEEL) : FLX9 * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

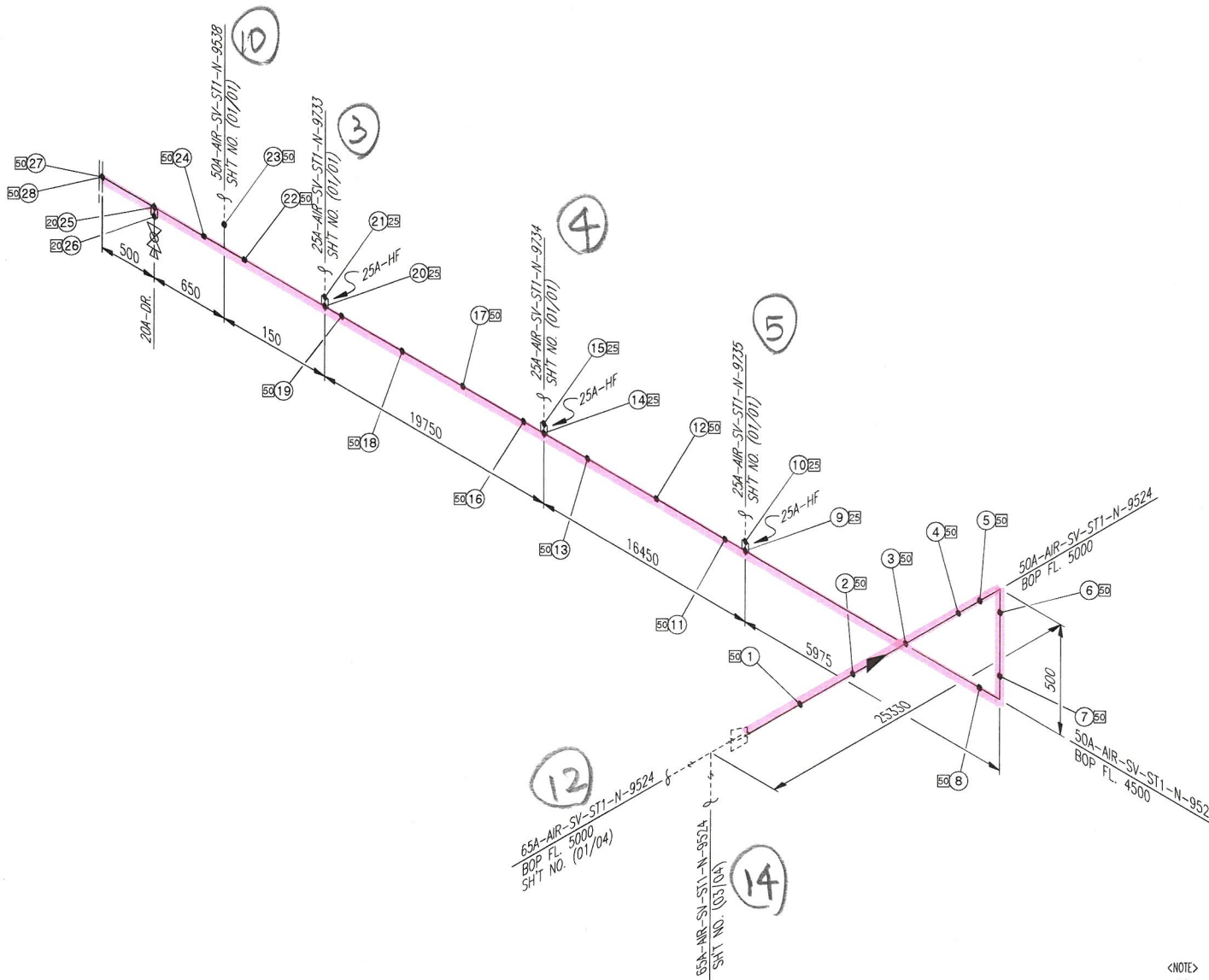
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14
	THICK.	-	DESIGN	10	60	UNIT	4FA
			OPERATING	7.5	AMB	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-040008-A	
						REF. P & ID	
						CJ-M15W13-PR-A20-000083-A	

CLIENT	SK hynix	PROJECT NAME	M15X Ph4 Project
CONTRACTOR	SK ecoplant	JOB NO.	223300D
COMMODITY	AIR-SV	SH'T NO.	01 / 04
LINE NO.	65A-AIR-SV-ST1-N-9524	REV.	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.

[illegible]

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생한 경우 유표자에게 민·형사상 책임을 요구할 수 있음.



SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	50A	69.3M	PIPE, STS 304TP, ERW, BE - SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, TDE - SCH 40S
E90	50A	2EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	50A	1EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
HF	25A	3EA	COUPLING HALF, STSF304, SW - SCH 10S
HF	20A	1EA	COUPLING HALF, STSF304, SW - SCH 10S
PLG	20A	1EA	PLUG ROUND HEAD, STSF304, SCR'D
VB1	20A	1EA	BALL VALVE, KS 10K, SCR'D, BODY-STSF304 / TRIM-STSF304 / SEAT-PTFE, FB, LEVER
F	50A	1EA	FLG SO, KS 10K, RF, STS304
FB	50A	1EA	FLG BLIND, KS 10K, RF, STS304
G	50A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	50A	1EA	MACHINE BOLT & NUTS : STS 304 / W WASHERS

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SOC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
 * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	4FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-040008-A	REF. P & ID	CJ-M15W13-PR-A20-000083-A	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	10	60										25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
			OPERATING	7.5	AMB														

CLIENT	SK hynix	15
CONTRACTOR	SK ecoplant	
COMMODITY	AIR-SV	PROJECT NAME M15X Ph4 Project
LINE NO.	50A-AIR-SV-ST1-N-9524	JOB NO. 22330D
		SH'T NO. 04 / 04
		REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.

PUNCH LIST

Project No : 22330D

Package No :

Project Title : M15X Ph3 Project

Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning